THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A sector of a circle of radius 42 cm has central angle 75°. The perimeter of the sector is: 1

(a) 97 cm **(b)** 139 cm **(c)** 55 cm **(d)** 181 cm

2. An arc of length 44 cm subtends an angle at the centre of a circle of radius 42 cm. The angle subtended by the arc at the centre is: 1

(a) 30° **(b)** 90° **(c)** 60° **(d)** 120°

3. A square is inscribed in a circle of radius 42 cm. The area of the circle outside the square is: 1

(a) 3780 cm² **(b)** 5544 cm² **(c)** 2016 cm² **(d)** 3528 cm²

4. A horse is tied at a corner of a large square field. The rope is first 21 m long and then lengthened to 35 m. The increase in the area it can graze is: 1

(a) 2464 m² **(b)** 154 m² **(c)** 962.5 m² **(d)** 616 m²

5. A chord of a circle of radius 17.5 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) The area of the minor segment is: 1

(a) 292.87 cm² **(b)** 160.42 cm² **(c)** 132.45 cm² **(d)** 27.96 cm²

6. A brooch is a circular piece of silver wire of diameter 14 mm. The wire is also used for 3 equal diameters that divide the brooch into 6 sectors. The total length of wire required (in mm) is: 1

(a) 65 mm **(b)** 154 mm **(c)** 44 mm **(d)** 86 mm

7. A horse is tied at the midpoint of one side of a large square field with a rope 49 m long. The area of the field it can graze is: 1

(a) 3773 m² **(b)** 7546 m² **(c)** 77 m² **(d)** 154 m²

8. A sector of a circle of radius 42 cm has central angle 135°. The perimeter of the sector is: 1

(a) 225 cm **(b)** 183 cm **(c)** 141 cm **(d)** 99 cm

9. **Assertion (A):** If the radius of a circle is doubled, its area also becomes double. 1

Reason (R): The area of a circle of radius r is πr^2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The area of a circle of diameter 14 cm is $154\ \text{cm}^2$ (use $\pi = \frac{22}{7}$). 1
Reason (R): The area of a sector of a circle is less than the area of the circle.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A wheel has radius 49 cm. Find the number of complete revolutions it makes to cover 154 m. 2

12. A circular park of radius 45 m is surrounded by a path 7 m wide. Find the area of the path. 2

13. An umbrella has 20 ribs which are equally spaced and the umbrella, when opened flat, is a circle of radius 70 cm. Find the area between two consecutive ribs. 2

14. A wheel has radius 56 cm. Find the number of complete revolutions it makes to cover 176 m. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A sector of a circle of radius 42 cm has central angle 110°. Find the area of the sector. 3

16. A quadrant of a circle of radius 77 cm is cut from a square of side 77 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

17. The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 20 minutes. 3

OR

A sector of a circle of radius 42 cm has central angle 140°. Find the area of the sector. 3

18. A quadrant of a circle of radius 84 cm is cut from a square of side 84 cm, with the centre of the quadrant at one corner of the square. Find the area of the remaining part of the square. 3

OR

The minute hand of a clock is 42 cm long. Find the area swept by the minute hand in 25 minutes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A decorative circular window has radius 14 cm. A straight metal strip is fixed along a chord AB that subtends a right angle at the centre O. The region between the chord and the minor arc is filled with coloured glass. (Take $\pi = \frac{22}{7}$.) 5
(i) Find the area of the sector OAB.
(ii) Find the area of $\triangle OAB$.
(iii) Find the area of the coloured (minor) segment.
(iv) Find the area of the major segment.

20. A chord of a circle of radius 56 cm subtends an angle of 120° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

OR

Two concentric circles have radii 21 cm and 14 cm. A sector of angle 108° is marked in both. Find the area of the region between the two arcs within the sector. 5

END OF QUESTION PAPER — **Total: 40 Marks**